# Lección 7

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06/04/2026

# Plan de trabajo

**Formulación Lagrangiana**

– Teorema de Noether: Simetrías→Integrales de movimiento

– Similtud Mecánica. Teorema de Virial

– Aplicaciones: movimiento en campo central

**Nuestro Syllabus:**

**FIS-310**:

UNIDAD 2. Formulación Lagrangiana
2.1. Variables Generalizadas
2.2. Clasificación de los Sistemas Mecánicos
2.3. Elementos de Cálculo Variacional
2.4. Ecuaciones de Lagrange
2.5. Aplicaciones: movimiento en un campo central, observadores no inerciales

UNIDAD 3. Teoremas de Conservación
3.1. Teorema de Noether.
3.2. Aplicaciones del teorema de Noether.

**FIS-345**:

Unidad I: Principios Variacionales, Lagrangianos y Hamiltonianos

- Principio de Hamilton y Lagrangianos
- Simetrías y Leyes de Conservación (Teorema de Noether)

Unidad II: Algunas Aplicaciones

- Movimiento en campo de Fuerzas Centrales
- Movimiento de Sólido Rígido

**FIS-426**:

1) Principios Variacionales, Lagrangianos y Hamiltonianos

- Principio de Hamilton y Lagrangianos
- Hamiltonianos
- Otros principios variacionales
- Simetrías y Leyes de Conservación (Teorema de Noether)

# Summary of the last lecture: Noether's theorem

For any continuous symmetry group of the action $\mathcal{S}$, there is a conserved quantity (integral of motion).

Assume symmetry transformation which mixes coordinates and time:

$$\boldsymbol{q} \to \boldsymbol{q}_\epsilon \approx \boldsymbol{q} + \epsilon\,\boldsymbol{\gamma}, \quad t \to t_\epsilon \approx t + \epsilon\,\tau, \qquad \epsilon \ll 1, \qquad \epsilon = \mathrm{const}$$

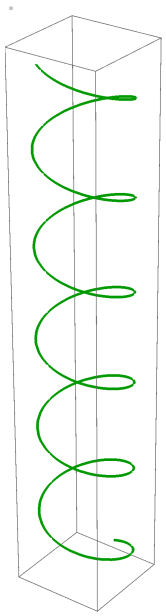
–Constants in front of $\epsilon$ (namely $\boldsymbol{\gamma}$, $\tau$) can be functions of coordinates, time
–We assume now that $\epsilon = \epsilon(t)$, i.e. acquires a small $t$-dependence.
1) Make Taylor expansion of $S$, picking up the linear $\mathcal{O}(\epsilon)$, $\mathcal{O}(\dot{\epsilon})$ terms

$$S_\epsilon = \int dt_\epsilon\, L\left(q_\epsilon, \dot{q}_\epsilon, t\right) \approx S + \int dt \left[\dot{\epsilon}\, Q + \epsilon \underbrace{(\ldots)}_{=0}\right]$$

– Term $\sim \epsilon$ vanishes because for $\epsilon = \mathrm{const}$ action $S = \mathrm{invariant}$
–General result for invariant $Q$:

$$\Rightarrow Q = \lim_{\varepsilon \to 0} \frac{\partial L_\epsilon\left(q_\epsilon, \dot{q}_\epsilon, t\right)}{\partial \dot{\epsilon}}, \quad L_\epsilon\left(q_\epsilon, \dot{q}_\epsilon, t\right) = \left(1 + \frac{\partial t_\epsilon}{\partial t}\right) L\left(q_\epsilon, \dot{q}_\epsilon, t\right)$$

# Beware: only symmetries of $S$ matter (not only $U$)



▷ Action of a particle in external field:

$$S = \int dt\, L = \int dt \left[ \frac{mv^2}{2} - U \right]$$

$$v^2 = g_{ij} v^i v^j$$

**Compare 2 problems:**

Assume that a charged particle moves in the external potential, which in cylindric coordinates is given by $U = U(r, z - a\phi)$. The potential has a continuous symmetry w.r.t.

$$z \to z + \epsilon,\ \phi \to \phi + \frac{\epsilon}{a} \tag{1}$$

Metrics in cylindr. coords: $g_{ij} = \{1, r^2, 1\}$ is invariant w.r.t. (1)
⇒There is integral of motion (Noether invariant) associated with symmetry (1) (which might be found taking derivative $\partial L / \partial \dot{\epsilon}$)

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Assume that a charged particle moves in the external potential, which in cylindrical coordinates is given by $U = U(z - a\,r)$. The potential has a continuous symmetry w.r.t.

$$z \to z + \epsilon,\ r \to r + \frac{\epsilon}{a} \tag{2}$$

Metrics $g_{ij} = \{1, r^2, 1\}$ is NOT invariant w.r.t. (2)
⇒Kinetic energy is NOT invariant w.r.t symmetry (2)
⇒There is NO integrals of motion (Noether invariants) associated with symmetry (2)

# Some important examples

"Classical" invariants ($\boldsymbol{n} = \mathrm{const}$ below):

1) Translations $\boldsymbol{q} \to \boldsymbol{q} + \epsilon \boldsymbol{n} \Rightarrow$ momentum projection $\boldsymbol{p} \cdot \boldsymbol{n}$

2) Rotations $\boldsymbol{q} \to \boldsymbol{q} + \epsilon \boldsymbol{n} \times \boldsymbol{q} \Rightarrow$ angular momentum projection $\boldsymbol{p} \cdot \boldsymbol{n}$

3) Less trivial example:

−For the nonrelativistic particle in the field

$$L = \frac{m\,\dot{\boldsymbol{r}}^2}{2} - U(x, y, z - V\,t)$$

there is a symmetry

$$z \to z + V\,\epsilon,\ t \to t + \epsilon, \qquad \epsilon = \mathrm{const}$$

−The Noether's invariant:

$$Q = \left.\frac{\partial L_\epsilon}{\partial \dot{\epsilon}}\right|_{\varepsilon \to 0} = L - \dot{\boldsymbol{r}}_a \frac{\partial L}{\partial \dot{\boldsymbol{r}}_a} + V\,\hat{\boldsymbol{z}}\,\frac{\partial L}{\partial \dot{z}} = -E(t) + p_z(t)V = -E_{\mathrm{r.f.}} + \frac{mV^2}{2}$$

where $E_{\mathrm{r.f.}}$ is the energy in the "co-moving frame", where $z' = z - Vt$ ant $U(\boldsymbol{r})$ does not depend on time

# Example 2

Demonstrate that the action $S$ of a free particle is invariant with respect to rescaling transformation

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \quad t \to \alpha^2 t.$$

Construct the integral of motion which corresponds to this symmetry according to the Noether's theorem. Demonstrate that this quantity is integral of motion using the equations of motion.

***Help***: Instead of variable $\alpha$ You may find easier to work with variable $\epsilon \equiv \ln \alpha$ (so $\alpha = e^{\epsilon} \approx 1 + \epsilon$ for $|\epsilon| \ll 1$).

**Important observation** (comment on common error of students):

−We always assumed that the identical transformation corresponds to parameter $\epsilon = 0$

−Here we **can't** make expansion over $\alpha$ directly, assuming $\alpha \ll 1$

*__Reason__: identity transformation corresponds to $\alpha = 1$, we should expand near it.

⇒We MUST make some (any) change of variables, like $\alpha = e^{\epsilon} \approx 1 + \epsilon$, so that trivial (identity) transformation is $\epsilon = 0$, and only after this make expansion over $\epsilon$ when we derive the Noether's invariant

# Example 2 (proof of invariance)

$$S = \int dt \, \frac{m \, \dot{\boldsymbol{r}}^2}{2}$$

Change of different quantities under rescaling:

$$\boldsymbol{r} \to \alpha \, \boldsymbol{r}, \quad t \to \alpha^2 t, \qquad \alpha = e^{\epsilon} \approx 1 + \epsilon \qquad (1)$$

$$\dot{\boldsymbol{r}} \equiv \frac{d\boldsymbol{r}}{dt} \to \frac{1}{\alpha} \dot{\boldsymbol{r}} \approx \dot{\boldsymbol{r}} - \epsilon \, \dot{\boldsymbol{r}},$$

$$\frac{m \, \dot{\boldsymbol{r}}^2}{2} \to \frac{1}{\alpha^2} \frac{m \, \dot{\boldsymbol{r}}^2}{2} \approx \frac{m \, \dot{\boldsymbol{r}}^2}{2} - 2\epsilon \, \frac{m \, \dot{\boldsymbol{r}}^2}{2}$$

$$\Rightarrow S = \int dt \, L \to \int \cancel{\alpha^2} dt \, \frac{1}{\cancel{\alpha^2}} \frac{m \, \dot{\boldsymbol{r}}^2}{2} = \mathrm{const}$$

This proves that (1) is indeed a continuous symmetry of the action.

# Example 2 (derivation of invariant)

►consider $t$-dependent $\alpha = \alpha(t) = e^{\epsilon(t)}$, $|\epsilon| \ll 1$

$$d\boldsymbol{r} \to (d\boldsymbol{r} + \boldsymbol{r} d\epsilon) = (d\boldsymbol{r} + \boldsymbol{r}\,\dot{\epsilon} dt)\,, \qquad dt \to (1 + 2\dot{\epsilon} t)\, dt$$

$$\boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt} \to \frac{\boldsymbol{v} + \boldsymbol{r}\,\dot{\epsilon}}{1 + 2\dot{\epsilon} t} \approx \boldsymbol{v} + \dot{\epsilon}\,(\boldsymbol{r} - 2t\,\boldsymbol{v})$$

$$S \equiv \int dt\, L \to \int dt\, L\,(1 + 2\dot{\epsilon} t) + \int dt\, m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v})\,\dot{\epsilon}$$

$$= \int dt\, [m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2tL\,]\,\dot{\epsilon}$$

$$Q = \frac{\delta S}{\delta \dot{\epsilon}} = m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2tL = m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2t\left(\frac{m\,\boldsymbol{v}^2}{2}\right)$$

$$= m\,\boldsymbol{v} \cdot \boldsymbol{r} + 2t\left(-\frac{m\,\boldsymbol{v}^2}{2}\right) = \boldsymbol{p} \cdot \boldsymbol{r} - 2E\,t$$

▷we know that energy $\boldsymbol{E}$ and $\boldsymbol{p}$ are conserved for a free particle, and $\boldsymbol{p} = m\boldsymbol{v}$, $\boldsymbol{r} = \boldsymbol{v}t + \boldsymbol{r}_0$, so $Q = \boldsymbol{p} \cdot \boldsymbol{r}_0$ indeed must be integral of motion:

$$\frac{dQ}{dt} = \dot{\boldsymbol{p}} \cdot \boldsymbol{r} + \boldsymbol{p} \cdot \boldsymbol{v} - 2E = -\frac{\partial U}{\partial \boldsymbol{r}} \cdot \boldsymbol{r} + \boldsymbol{p} \cdot \boldsymbol{v} - 2E = 2\left(\frac{m\,\boldsymbol{v}^2}{2}\right) - 2E \equiv 0 \qquad \text{Q.E.D.}$$

# Example 3

Demonstrate that the action $S$ of a particle moving in the homogeneous potential $U(r) = -\kappa/r^2$ is invariant with respect to rescaling transformation

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \quad t \to \alpha^2 t$$

Construct the integral of motion which corresponds to this symmetry according to the Noether's theorem. Demonstrate that this quantity is conserving using the equations of motion.

***Help***: Instead of variable $\alpha$ You may find easier to work with variable $\epsilon \equiv \ln a$ (so $a \approx e^{\epsilon} \approx 1 + \epsilon$ for $|\epsilon| \ll 1$).

**Solution:**

$$S = \int dt \left( \frac{m\,\dot{\boldsymbol{r}}^2}{2} - \frac{\kappa}{r^2} \right)$$

Change of some quantities under rescaling:

$$\dot{\boldsymbol{r}} \equiv \frac{d\boldsymbol{r}}{dt} \to \frac{1}{\alpha}\dot{\boldsymbol{r}},$$

$$\frac{m\,\dot{\boldsymbol{r}}^2}{2} \to \frac{1}{\alpha^2}\frac{m\,\dot{\boldsymbol{r}}^2}{2}, \qquad U(r) \to \frac{1}{\alpha^2}U(r)$$

$$\Rightarrow S = \mathrm{const}$$

# Example 3 (derivation of invariant)

►consider $t$-dependent $\alpha = \alpha(t) = e^{\epsilon(t)}$, $|\epsilon| \ll 1$ and formally repeat what we did in Example 2:

$$d\boldsymbol{r} \to (d\boldsymbol{r} + \boldsymbol{r} d\epsilon) = (d\boldsymbol{r} + \boldsymbol{r}\,\dot{\epsilon} dt)\,, \qquad dt \to (1 + 2\dot{\epsilon} t)\, dt$$

$$\boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt} \to \frac{\boldsymbol{v} + \boldsymbol{r}\,\dot{\epsilon}}{1 + 2\dot{\epsilon} t} \approx \boldsymbol{v} + \dot{\epsilon}\,(\boldsymbol{r} - 2t\,\boldsymbol{v})$$

$$S \equiv \int dt\, L \to \int dt\, L\,(1 + 2\dot{\epsilon} t) + \int dt\, m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v})\,\dot{\epsilon}$$

$$= \int dt\, [m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2tL]\,\dot{\epsilon}$$

$$Q = \frac{\delta S}{\delta \dot{\epsilon}} = m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2tL = m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2t\left(\frac{m\,\boldsymbol{v}^2}{2} + \frac{\kappa}{r^2}\right)$$

$$= m\,\boldsymbol{v} \cdot \boldsymbol{r} + 2t\left(-\frac{m\,\boldsymbol{v}^2}{2} + \frac{\kappa}{r^2}\right) = \boldsymbol{p} \cdot \boldsymbol{r} - 2E\,t$$

Result for $Q$ looks similar to what we found in Example 2. However, now $E$ includes potential energy $U(r)$, and $\boldsymbol{p} \neq \mathrm{const}$

# Example 3 (proof of invariance)

$$Q = m\,\boldsymbol{v}\cdot\boldsymbol{r} + 2t\left(-\frac{m\,\boldsymbol{v}^2}{2} + \frac{\kappa}{r^2}\right) = \boldsymbol{p}\cdot\boldsymbol{r} - 2E\,t$$

Result for $Q$ looks similar to what we found in Example 2. However, now $E$ includes potential energy $U(r)$, and $\boldsymbol{p} \neq \mathrm{const}$

$-$Energy is conserved in time-independent potential so treat it as $\mathrm{const}$.

$-$Check that $Q = \mathrm{const}$:

$$\begin{aligned}\frac{dQ}{dt} &= \dot{\boldsymbol{p}}\cdot\boldsymbol{r} + \boldsymbol{p}\cdot\boldsymbol{v} - 2E \\ &= -\frac{\partial U}{\partial \boldsymbol{r}}\cdot\boldsymbol{r} + \boldsymbol{p}\cdot\boldsymbol{v} - 2E = \\ &= 2\left(-\frac{\kappa}{r^2} + \frac{m\,\boldsymbol{v}^2}{2}\right) - 2E \equiv 0\end{aligned}$$

as expected !!!

[Landau, Vol. 1, Section §10]

# Mechanical similarity

# Mechanical similarity (introduction)

[Landau, Vol. 1, Section §10]

– Previous problems: the action in the field $U = \kappa/r^2$ is invariant w.r.t. transformation,

$$\boldsymbol{r} \to \alpha\,\boldsymbol{r}, \quad t \to \alpha^2 t, \qquad (1)$$

i.e. if $\boldsymbol{r}(t)$ is classical trajectory, then $\alpha\boldsymbol{r}\left(\alpha^2\, t\right)$ is also a classical trajectory.

– Assume that now we consider the *homogeneous* field $U = \kappa\, r^k$, where $k$ may be any real number.

We may expect that there are also transformations of type

$$\boldsymbol{r} \to \alpha\,\boldsymbol{r}, \quad t \to \beta t, \qquad (2)$$

such that if $\boldsymbol{r}(t)$ is a classical trajectory, then $\alpha\boldsymbol{r}\,(\beta\, t)$ is also a classical trajectory.

# Mechanical similarity (introduction)

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \quad t \to \beta t, \qquad (2)$$

– If we'll try to fix $\alpha, \beta$ from condition $S = \mathrm{const}$, we'll find that it is impossible, since kinetic and potential energy transform differently

$$S \to \frac{\alpha^2}{\beta} \int dt\, \frac{mv^2}{2} - \beta\, \alpha^k \int dt\, U$$

(only for $\beta = \alpha^2$ **and** $k = -2$ they both coincide).

– So we'll use another method and show that "scaling law" for trajectories has a form (2) with $\beta = \alpha^{1-k/2}$, and the action transforms as

$$S \to \alpha^{1+k/2} S$$

Under transformation (2), for $k \neq 2$ the action is NOT invariant, $S \neq \mathrm{const}$, so **the Noether's theorem is not applicable**, and (apparently) there is no associated invariants.

# Lemma

Assume that the potential energy is a homogeneous function of coordinates, viz:

$$U(\alpha \boldsymbol{r}_1, \ldots, \alpha \boldsymbol{r}_n) = \alpha^k U(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_n) \qquad (1)$$

where $k$ is some number. If $\{\boldsymbol{r}_i(t)\}$ is a solution of the EL equations, then $\{\alpha \boldsymbol{r}_i(\beta t)\}$ is also a solution of the problem, where $\beta = \alpha^{1-k/2}$.

**Proof**: Let's analyze how different parts of EL equation transform under rescaling (1):

$$L = \sum_k \frac{m_k \dot{\boldsymbol{r}}_k^2}{2} - U(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_k) \to \left(\frac{\alpha}{\beta}\right)^2 \sum_k \frac{m_k \dot{\boldsymbol{r}}_k^2}{2} - \alpha^k U(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_k)$$

$$= \alpha^k \left[ \left(\frac{\alpha^{2-k}}{\beta^2}\right) \sum_k \frac{m_k \dot{\boldsymbol{r}}_k^2}{2} - U(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_k) \right]$$

►If $\beta = \alpha^{1-k/2}$, the factor in red $\equiv 1$, so we just have rescaling of the lagrangian.

... – EL equations in this case are the same as for $\alpha = \beta = 1$, since common factor $\alpha^k$ appears in front of all terms in Euler-Lagrange equations and eventually cancels

*Explicit transformation of EL equations:

$$m\ddot{\boldsymbol{r}} \to \frac{\alpha}{\beta^2} m\ddot{\boldsymbol{r}}, \quad \nabla U \to \alpha^{k-1} \nabla U, \quad \frac{\alpha}{\beta^2} = \alpha^{k-1} \text{ for } \beta = \alpha^{1-k/2}$$

# Mechanical similarity

**Special cases**:

- Case $k = -2$: $\beta = \alpha^2$, we recover that in potential $U = \kappa/r^2$ the trajectory is invariant w.r.t. transformation

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \quad t \to \alpha^2 t, \qquad (1)$$

- Case $k = 2$: $\beta$ does not depend on $\alpha$, period of oscillation does not depend on energy (harmonic oscillator)
- Case $k = -1$: $\beta \sim \alpha^{3/2}$, for closed orbits have Kepler's 3rd law ($T^2 \sim a^3$, where $T$ is period and $a$ is the major semiaxis)

*You might have seen in undergraduate courses that these properties are valid for a single particle in harmonic oscillator and Coulomb potential. Now we claim that they are valid for a system of particles interacting with each other via homogeneous potential*

[Landau, Vol. 1, Section §10]
[Goldstein, Section §3.4]

# Virial theorem

# Virial theorem

[Landau, Vol. 1, Section §10]
[Goldstein, Section §3.4]

—Sometimes (e.g. in thermodynamics) we may have sopshiticated system which cannot be integrated analytically, but we need relations between *average values* of the kinetic, potential energy, etc.

—There is a theorem called **virial theorem** which allows to solve this problem ...

# Virial theorem

Let's assume that the system moves over a closed trajectory in a potential which is a homogeneous function of coordinates, viz:

$$U(\alpha \boldsymbol{r}_1, \ldots, \alpha \boldsymbol{r}_n) = \alpha^k U(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_n) \qquad (1)$$

Then the averaged (over the period of motion) kinetic ($T$) and potential ($U$) energies satisfy

$$2\bar{T} = k\,\bar{U}$$

where we introduced "average" of observable $A$ as

$$\forall A, \quad \bar{A} \equiv \frac{1}{T}\int_0^T dt\, A(\boldsymbol{r}(t),\, t)$$

**Proof**:

$$2T = \sum p \cdot v = \frac{d}{dt}(\boldsymbol{p}\cdot\boldsymbol{r}) - \sum \dot{\boldsymbol{p}}\cdot\boldsymbol{r}$$

- After averaging over the period of motion the first term in rhs does not contribute.
- The second term due to EOM gives $\sum \boldsymbol{r}\cdot\boldsymbol{\nabla} U$

...—For homogeneous function $\int dt \sum \boldsymbol{r}\cdot\boldsymbol{\nabla} U \sim k \int dt\, U$, so this proves the statement of virial theorem

# Corollary

$$2\bar{T} = k\,\bar{U}$$

►Since energy is conserved for conservative forces, $E = \bar{T} + \bar{U}$, so

$$\bar{U} = \frac{2}{k+2}E, \quad \bar{T} = \frac{k}{k+2}E$$

–For $k = 2$ (harmonic oscillator) $\overline{T} = \overline{U}$
–For $k = -1$ (Kepler problem) $\overline{T} = -E, \ \overline{U} = 2E$

## Problem

What happens for $k = -2$ (potential $U = -\kappa/r^2$)? Can You explain this ?

[Landau, Vol. 1, Section §10]
[Goldstein, Section §3.4]

# Motion in central field

# Motion in central field (3D)

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Assume that a massive particle moves in the central field $U(r)$.

Please try to recall as much as You can about the main qualitative features of the motion ...

Some warm-up questions ...

– Identify all integrals of motion in this problem

– Analyze how existence of these integrals affects the trajectories

– What can You say about special case when

$$U(r) = -\alpha/r$$

(Kepler problem) ?

– Assume now that the potential is central BUT depends on time, $U = U(r, t)$. Which of your previous replies remain valid in this case ?

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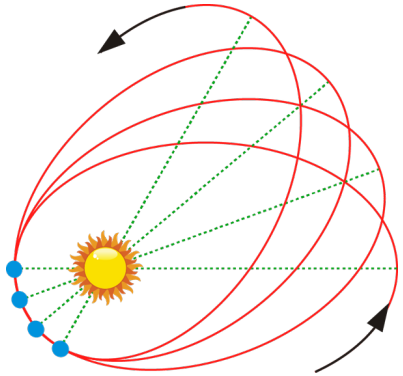
# Introductory questions ...

Assume that a particle moves in the central time-independent potential $U(r)$

– How to find the trajectory $r(\varphi)$ or $r(t)$ explicitly ?



– Is the motion periodic in general case?

  * In what potentials the trajectories are always periodic (for *any* trajectory) ?

– Can the particle fall on center (reach the point $r = 0$) ?

  * In what potentials the particle will *never* fall on center?

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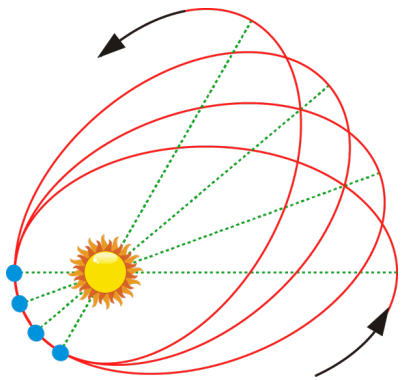

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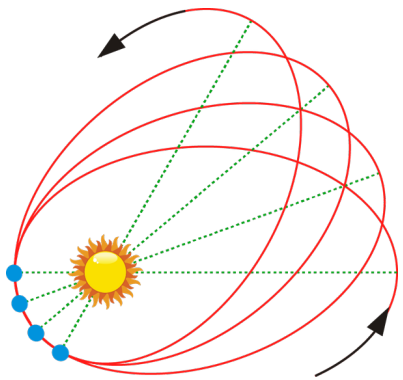


# Central field: Lagrangian and Euler-Lagrange equations

$$L = T - U = \frac{m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\varphi}^2\right)}{2} - U(r)$$

- Explicit Euler-Lagrange equations:

$$m\ddot{r} \equiv m\frac{d^2 r}{dt^2} = -\frac{\partial U}{\partial r} + mr\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\varphi}^2\right),$$

$$\frac{d(m\,r^2\dot{\theta})}{dt} \equiv m\,r^2\ddot{\theta} + 2mr\,\dot{r}\dot{\theta} = \frac{m\,r^2\sin 2\theta\,\dot{\varphi}^2}{2}$$

$$\frac{d(m\,r^2\sin^2\theta\dot{\varphi})}{dt} = m\,r^2\sin^2\theta\ddot{\varphi} + m\,r^2\sin 2\theta\,\dot{\theta}\dot{\varphi} + 2m\,r\,\dot{r}\dot{\varphi}\sin^2\theta = 0$$

–too complicated for direct solution
–Force $(-\partial U/\partial r)$ appears only in the first equation
–use integrals of motion to simplify

# Integrals of motion in central field (3D)

- Equations of Euler-Lagrange (equations of motion) are complicated nonlinear 2nd order ODEs, not easy to integrate them directly
  - As usual, can try to use conservation laws to find the solution
    - **Energy conservation:

$$E = \sum_a \dot{q}_a \frac{\partial L}{\partial \dot{q}_a} - L = \frac{m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 \sin^2\theta\, \dot{\varphi}^2\right)}{2} + U(r) = \text{const}$$

    - **Angular momentum conservation:

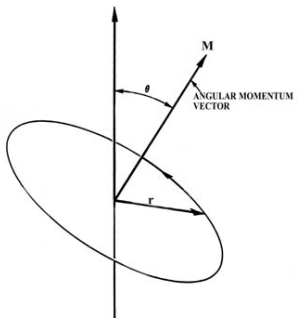
$$\boldsymbol{M} = \boldsymbol{r} \times \boldsymbol{p} = r\,\boldsymbol{e}_r \times m\boldsymbol{v} = \left( \underbrace{-mr^2 \sin\theta\, \dot{\varphi}}_{M_\theta = -p_\varphi / \sin\theta} \boldsymbol{e}_\theta + \underbrace{mr^2\, \dot{\theta}}_{M_\varphi = p_\theta} \boldsymbol{e}_\varphi \right) = \text{const}$$

      - ***Cartesian coord.: $M_{x,y,z} = \text{const}$
      - *** Spherical coord.: $M_{\theta,\varphi} \neq \text{const}$ because $\boldsymbol{e}_\theta$, $\boldsymbol{e}_\varphi$ change orientation from point to point.
      - ***Absolute value: $M^2 = \boldsymbol{M} \cdot \boldsymbol{M} = \text{const}$ in any ref. frame;

$$M^2 = m^2 r^4 \left(\dot{\theta}^2 + \sin^2\theta\, \dot{\varphi}^2\right) = \text{const}$$

# Consequence of angular momentum conservation

Since we have conservation of ALL components of vector, there is a remarkable property:




- *Particle* <u>*always*</u> *moves in a plane orthogonal to* $\boldsymbol{M}$*, i.e. if in the moment* $t_0$ *vectors* $(\boldsymbol{r}, \boldsymbol{v})$ *lie in a plane orthogonal to* $\boldsymbol{M}$*, then in the central field they remain in this plane for any moment* $t > t_0$ *in the future*:

**Proof**: We'll try to prove the statement assuming that the interval $\Delta t = t - t_0$ is small, and will make Taylor expansion over $\Delta t$:

$$\boldsymbol{r}(t_0 + \Delta t) \approx \boldsymbol{r}(t_0) + \Delta t\, \boldsymbol{v}, \quad \boldsymbol{r}\,(t_0) \cdot \boldsymbol{M} = 0$$
$$\boldsymbol{v}(t_0 + \Delta t) \approx \boldsymbol{v}(t_0) + \Delta t\, \boldsymbol{a}, \quad \boldsymbol{v}\,(t_0) \cdot \boldsymbol{M} = 0$$

$$\Delta t\, \boldsymbol{v} \cdot \boldsymbol{M} = \Delta t\, \boldsymbol{v} \cdot m\,(\boldsymbol{r} \times \boldsymbol{v}) = 0,$$
$$\Delta t\, \boldsymbol{a} \cdot \boldsymbol{M} = \frac{\Delta t\, \boldsymbol{F} \cdot \boldsymbol{M}}{m} = \frac{\Delta t\, \boldsymbol{r} \cdot \boldsymbol{M}}{m} \frac{dU}{r\, dr} = 0$$

$\Rightarrow \boldsymbol{r}(t_0 + \Delta t)$, $\boldsymbol{v}(t_0 + \Delta t)$ will be also orthogonal to $\boldsymbol{M}$.

– If we have finite interval $t - t_0$, we may split it into $N \gg 1$ small subintervals, and repeat the proof recursively for each piece.

# Consequence of angular momentum conservation (II)

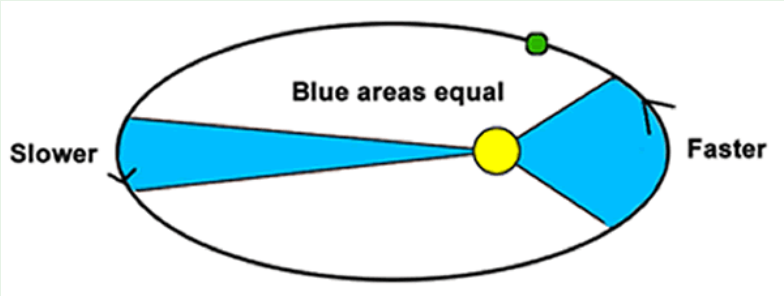
- In what follows we will select the $z$-axis of our reference frame along $\boldsymbol{M}$, so that the particle moves in the plane $z = 0$. In spherical coordinates this corresponds to $\theta = \frac{\pi}{2} = \mathrm{const}$, and thus $\dot{\theta}, \ddot{\theta} = 0$.

**Equation of azimuthal angle $\varphi$:**

$$M_z \equiv m\, r^2 \dot{\phi} \underbrace{\sin^2\theta}_{=1} = \mathrm{const} \quad \Rightarrow M\, dt = r^2 d\varphi = 2\, dS$$

►$dS = r^2 d\varphi/2$ is the area of line segment swept by the radius-vector $\boldsymbol{r}(t)$ per interval of time $dt$. So we now obtain a new law:

**Second law of Kepler**




A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time

- Discovered by observations of the planet motion (gravitational potential $U(r) \sim -\alpha/r$), approximately 50 years *before* the Newton's laws were formulated.
- Valid in *any* central field $U(r)$, not only in the Coulomb's field

# Integration of equations of motion

- Angular momentum (absolute value):

$$M^2 = m^2 r^4 \left(\dot{\theta}^2 + \sin^2\theta\,\dot{\varphi}^2\right) = \text{const} \qquad (1)$$

- Energy:

$$E = \frac{m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\varphi}^2\right)}{2} + U(r) = \frac{m\dot{r}^2}{2} + \underbrace{U(r) + \frac{M^2}{2mr^2}}_{U_{\text{eff}}(r)} \qquad (2)$$

–Effective motion in 1 dimension (radial), in the field of effective potential $U_{\text{eff}}(r)$
–The universal contribution $+M^2/\left(2\,m\,r^2\right)$ is called potential of the centrifugal forces; its derivative $\boldsymbol{F} = -\boldsymbol{\nabla} U$ gives additional centrif. force in the radial direction

**Formal solution:**

$$\frac{dr}{dt} = \sqrt{\frac{2}{m}\left[E - U_{\text{eff}}(r)\right]} \qquad \Rightarrow \int_{r_0}^{r} \frac{dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}} = t - t_0 \qquad (3)$$

$$M \equiv m\,r^2\dot{\varphi} = \text{const} \quad \varphi - \varphi_0 = \int_{\varphi_0}^{\varphi} d\phi = \int_{t_0}^{t} dt\,\frac{M}{mr^2(t)}. \qquad (4)$$

so if we extract $r(t)$ from (3), can use (4) to find the trajectory $\{r(t), \varphi(t)\}$.

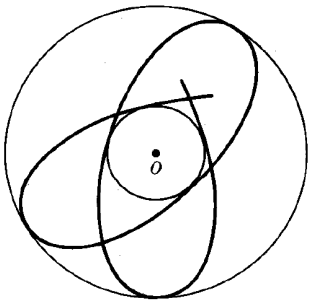
# Motion in central field - trajectory

Now let's evaluate the shape of trajectory *explicitly, without its time dependence:*

$$\frac{dr}{dt} = \sqrt{\frac{2}{m}\left[E - U_{\text{eff}}(r)\right]} \qquad (1)$$

$$M \equiv m\, r^2 \dot{\phi} = \text{const} \qquad (2)$$

- Let's assume that $r(\varphi)$ is the trajectory of the particle



$$\frac{dr}{d\varphi} = \frac{dr/dt}{d\varphi/dt} = \frac{\dot{r}}{\dot{\varphi}} = \frac{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}}{M/(mr^2)}$$

$$\Downarrow$$

$$\varphi - \varphi_0 = \int^r dr\, \frac{M}{m\, r^2 \sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}}$$

In general for finite motion the trajectories are not closed (not periodic)

# Mini-summary

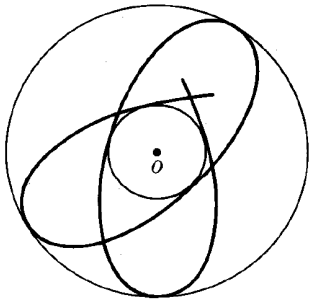
- Time dependence $\{r(t), \phi(t)\}$ might be extracted from:

$$\int_{r_0}^{r} \frac{dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}} = t - t_0, \quad U_{\text{eff}}(r) = U(r) + \frac{M^2}{2\,m\,r^2} \tag{1}$$

$$\varphi - \varphi_0 = \int^{r} dr \frac{M}{m\,r^2\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}} \tag{2}$$

- If we are NOT interested in time dependence, only in geometrical shape $r = r(\phi)$, can use expression

$$\phi - \phi_0 = \int_{r_0}^{r} dr \frac{M}{m\,r^2\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}} \tag{3}$$



In general for finite motion the trajectories might be not closed (not periodic).
–It might be periodic for some orbits, but not for all.
–Only for 2 potentials ($U = -\alpha/r$ and $U = kr^2$) it is periodic for any orbit.

- Equations (1-2) constitute main quantitative results which allow to describe trajectories in any field.